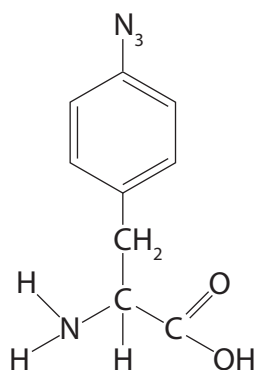


- 8 Silkworms are caterpillars that produce silk. Silk is a fibrous protein that can be used in clothing and in medicine.

Silkworms have been selectively bred to produce a modified silk that could have even more medical uses.

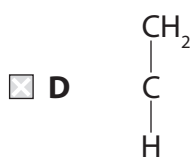
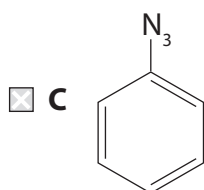
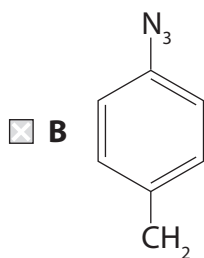
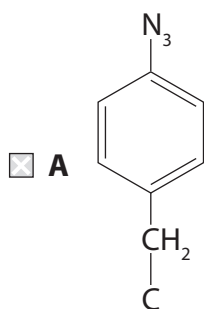
The modified silk is made by these silkworms by inserting a synthetic amino acid, AzPhe, into the protein. This replaces the naturally-occurring amino acid phenylalanine.

The diagram shows the structure of AzPhe.



- (a) Which of the following diagrams shows the R group only of AzPhe?

(1)



- (b) Only the silkworms that have been selectively bred can use AzPhe in the synthesis of protein.

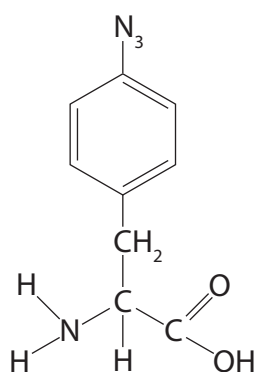
During translation, AzPhe joins to two other amino acids by condensation reactions.

- (i) Name the covalent bond that joins two amino acids in a condensation reaction.

(1)

- (ii) Draw two circles on the diagram of AzPhe to show which parts of the molecule are lost when AzPhe joins to two other amino acids by condensation reactions.

(2)



- (iii) Transfer RNA (tRNA) is involved in translation. The amino acid AzPhe requires a special tRNA molecule during the synthesis of silk.

Suggest why AzPhe is **not** inserted into proteins in silkworms that have not been selectively bred.

(3)



- (c) Part of a silk molecule contains 1100 amino acids. In natural silk, 1% of the amino acids are phenylalanine.

In modified silk, 16% of the phenylalanine molecules are replaced by AzPhe.

- (i) Calculate the ratio of phenylalanine to AzPhe in this part of a modified silk molecule.

(2)

Answer

- (ii) The R group of phenylalanine is smaller than the R group of AzPhe.

Suggest how inserting an amino acid with a larger R group could affect the properties of silk fibres.

(4)

(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS



P 6 1 4 6 9 A 0 2 7 2 8